

SIMATS ENGINEERING
ASSESSMENT TOOL 3: MCQ Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0509

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Assessment Tool Weightage: 10%

QUESTIONS: 50

Total Marks:50

Annexure A
MCQ Test

Code of Conduct:

I N. Akshaya (192425129) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Akshaya

Q.No	Question	Option A	Option B	Option C	Option D	Ans
1	Preprocessing improves feature detection by	A) Increasing size	B) Reducing noise	C) Changing format	D) Increasing brightness	B
2	Without preprocessing, detection becomes	A) Better	B) Same	C) Inconsistent	D) Faster	C
3	Grayscale conversion helps in	A) Increasing color	B) Reducing complexity	C) Increasing resolution	D) Adding noise	B
4	Binary representation is useful for	A) Texture detection	B) Foreground separation	C) Color enhancement	D) Contrast increase	B

Q.No	Question	Option A	Option B	Option C	Option D	Ans
5	Image representation affects	A) Pixel size	B) Feature visibility	C) Memory only	D) Edge removal	B
6	Edge detection identifies	A) Noise	B) Intensity change	C) Smooth regions	D) Colors	B
7	Canny is preferred due to	A) Speed	B) Low error	C) Memory	D) Size	B
8	Sobel is sensitive to	A) Color	B) Noise	C) Size	D) Resolution	B
9	Weak edges occur due to	A) High contrast	B) Low gradients	C) High resolution	D) Size	B
10	Edge detection fails when	A) Smooth image	B) Strong gradients	C) Noise dominates	D) High resolution	C
11	Corner detection finds	A) Flat regions	B) Edges	C) Intensity variation	D) Noise	C
12	Harris is sensitive to	A) Rotation	B) Translation	C) Intensity	D) Noise	C
13	Corners are useful because	A) Smooth	B) Distinctive	C) Large	D) Less intensity	B
14	Few corners due to	A) High contrast	B) Excess smoothing	C) High resolution	D) Sharp edges	B
15	Harris works well in	A) Uniform	B) Textured	C) Flat	D) Blurred	B
16	Hough Transform is for	A) Noise removal	B) Shape detection	C) Compression	D) Color	B
17	It works using	A) Ignoring noise	B) Voting	C) Reducing pixels	D) Grayscale	B
18	Poor Hough result due to	A) Clean image	B) Noise	C) Strong edges	D) High resolution	B
19	Hough maps to	A) Frequency	B) Parameter space	C) Colors	D) Edges	B
20	Preprocessing helps Hough by	A) Adding noise	B) Enhancing edges	C) Reducing resolution	D) Removing features	B
21	SIFT is invariant to	A) Noise	B) Scale & rotation	C) Color	D) Resolution	B
22	Matching fails when	A) Similar images	B) Weak features	C) Low noise	D) High resolution	B
23	SIFT works better due to	A) Color	B) Scale invariance	C) Size	D) Grayscale	B
24	Feature matching is used in	A) Compression	B) Recognition	C) Noise removal	D) Histogram	B

Q.No	Question	Option A	Option B	Option C	Option D	Ans
25	Incorrect matches due to	A) Strong features	B) Similar descriptors	C) High contrast	D) High resolution	B
26	PCA is used for	A) Edge detection	B) Dimensionality reduction	C) Noise addition	D) Smoothing	B
27	PCA removes	A) Features	B) Correlation	C) Noise	D) Pixels	B
28	PCA improves	A) Data size	B) Efficiency	C) Noise	D) Pixels	B
29	PCA may cause	A) Data loss	B) Noise	C) Edge loss	D) Color change	A
30	PCA is applied after	A) Segmentation	B) Feature extraction	C) Filtering	D) Thresholding	B
31	Preprocessing + features improve	A) Noise	B) Accuracy	C) Size	D) Color	B
32	Edge + Hough improves	A) Noise	B) Shape detection	C) Color	D) Compression	B
33	Corner detection used for	A) Noise removal	B) Matching	C) Smoothing	D) Thresholding	B
34	Feature extraction fails when	A) Noisy image	B) Clear image	C) Large image	D) Small image	A
35	Preprocessing improves	A) Noise	B) Features	C) Pixels	D) Edges	B
36	First step if edges fail	A) PCA	B) Preprocessing	C) SIFT	D) Hough	B
37	Scale issue solved by	A) Sobel	B) SIFT	C) PCA	D) Thresholding	B
38	Redundant features handled by	A) Edge	B) PCA	C) Hough	D) Smoothing	B
39	Missing corners fixed by	A) Threshold	B) Size	C) Color	D) Resolution	A
40	False Hough detection fixed by	A) Noise	B) Preprocessing	C) Color	D) Resolution	B
41	Shape detection uses	A) PCA+SIFT	B) Edge+Hough	C) Smooth+PCA	D) Hist+SIFT	B
42	Rotation matching uses	A) Sobel	B) SIFT	C) Threshold	D) PCA	B
43	Feature extraction helps	A) Compression	B) Key points	C) Noise	D) Color	B
44	Harris fails in	A) Textured	B) Uniform	C) Noisy	D) High contrast	B

Q.No	Question	Option A	Option B	Option C	Option D	Ans
45	Preprocessing needed for	A) Visualization	B) Feature extraction	C) Compression	D) Storage	B
46	Edge output used in	A) PCA	B) Hough	C) Smoothing	D) Threshold	B
47	PCA transforms to	A) Spatial	B) Frequency	C) Feature space	D) Binary	C
48	SIFT uses	A) Color	B) Gradients	C) Resolution	D) Size	B
49	Matching depends on	A) Size	B) Distinctiveness	C) Color	D) Resolution	B
50	Feature pipeline improves	A) Size	B) Accuracy	C) Noise	D) Compression	B